

NETFLIX DATA ANALYSIS REPORT

Prepared By

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1. Project Overview

Netflix is one of the world's largest online streaming platforms, offering thousands of movies and television shows across different genres and countries. With such a large collection of content, data analysis helps understand viewing trends, production patterns, content distribution, and overall business insights.

This project performs **Exploratory Data Analysis (EDA)** on the Netflix Titles Dataset using Python. The project focuses on cleaning the dataset, handling missing values, creating new features, visualizing important trends, and extracting meaningful insights from the data.

2. Project Objectives

The objectives of this project are:

- Understand the Netflix dataset.
 - Perform data cleaning and preprocessing.
 - Handle missing values.
 - Remove duplicate records.
 - Create useful features from existing columns.
 - Compare Movies and TV Shows.
 - Analyze content by country.
 - Analyze content by genre.
 - Identify the most featured directors.
 - Identify the actors appearing most frequently.
 - Analyze content ratings.
 - Study movie duration and TV show seasons.
 - Visualize important trends.
 - Generate meaningful business insights.
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3. Dataset Description

The Netflix Titles Dataset contains information about movies and television shows available on Netflix.

The important columns include:

- Show ID

- Content Type
- Title
- Director
- Cast
- Country
- Date Added
- Release Year
- Rating
- Duration
- Genre (Listed In)
- Description

This dataset contains thousands of Netflix titles collected from different countries around the world.

4. Tools and Technologies Used

Programming Language:

- Python

Libraries:

- Pandas
- NumPy
- Matplotlib
- Seaborn

Development Environment:

- Visual Studio Code
- Jupyter Notebook

Version Control:

- Git
 - GitHub
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5. Project Structure

```
Netflix_Data_Analysis/
```

```
data/  
images/  
notebooks/  
reports/  
README.md  
requirements.txt  
.gitignore
```

6. Data Collection

The dataset was downloaded from Kaggle and imported into Python using Pandas.

The following functions were used to inspect the dataset:

- `head()`
- `shape`
- `columns`
- `info()`
- `describe()`

These functions helped understand the dataset before starting the analysis.

7. Data Cleaning

Data cleaning was an important step before performing analysis.

The following operations were performed:

- Removed duplicate records.
- Filled missing values in the Director column with "Unknown".
- Filled missing values in the Cast column with "Unknown".
- Filled missing values in the Country column with "Unknown".
- Filled missing values in the Rating column with "Not Rated".
- Removed rows containing missing values in the Date Added column.
- Converted the Date Added column into DateTime format.

These steps improved data quality and ensured accurate analysis.

8. Feature Engineering

New features were created from the existing columns to simplify analysis.

From the Date Added column:

- Year Added
- Month Added
- Day Added
- Weekday Added

From the Duration column:

- Duration Value
- Duration Unit

Example:

90 min



Duration Value = 90

Duration Unit = min

These new columns made it easier to analyze movies and TV shows separately.

9. Exploratory Data Analysis

Several analyses were performed to understand the Netflix dataset.

The following visualizations were created:

- Missing Values Heatmap
- Movies vs TV Shows Distribution
- Content Added by Year
- Content Added by Month
- Top 10 Countries
- Top 10 Genres
- Top 10 Directors
- Top 10 Actors
- Rating Distribution
- Movie Duration Distribution
- TV Show Seasons Distribution
- Movies vs TV Shows Pie Chart
- Release Year Trend

These visualizations helped identify important trends and patterns in the dataset.

10. Key Business Insights

The following insights were obtained from the analysis:

- Movies represent a larger portion of Netflix's catalog than TV Shows.
- The United States contributes the highest number of Netflix titles.
- India is among the leading content-producing countries.
- Drama and International Movies are the most common genres.
- TV-MA is one of the most frequently assigned content ratings.
- Netflix experienced rapid content growth after 2015.
- Several directors have contributed multiple titles to Netflix.
- Some actors appear in numerous Netflix productions.
- Most movies have a duration between 80 and 120 minutes.
- Most TV Shows consist of only one season.

These findings provide valuable insights into Netflix's content strategy and growth.

11. Challenges Faced

Several challenges were encountered during the project.

- Handling missing values.
- Converting Date Added into DateTime format.
- Splitting the Duration column into value and unit.
- Handling NaN values while converting Duration Value to integers.
- Counting individual actors and directors from comma-separated strings.
- Saving charts automatically into the Images folder.

Each challenge was resolved using appropriate Pandas functions and data preprocessing techniques.

12. Conclusion

This project successfully demonstrated the complete Exploratory Data Analysis process using Python.

The project included:

- Data Loading
- Data Cleaning
- Feature Engineering
- Data Visualization
- Business Insight Generation

The analysis provided valuable insights into Netflix's content library and strengthened practical skills in data analysis using Python.

13. Future Improvements

The project can be enhanced by implementing the following features:

- Interactive dashboard using Plotly.
 - Business dashboard using Power BI.
 - Movie Recommendation System.
 - Sentiment Analysis on movie descriptions.
 - Machine Learning models for prediction.
 - Deployment using Streamlit or Flask.
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14. References

- Netflix Titles Dataset (Kaggle)
- Pandas Documentation

- [NumPy Documentation](#)
 - [Matplotlib Documentation](#)
 - [Seaborn Documentation](#)
 - [Python Official Documentation](#)
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15. Author

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